Everything you need to know about the new F1 rules for 2026

Formula 1 prides itself on being at the forefront of innovation – and the relentless pursuit of breaking new ground continues next year when the championship's rules will be revamped in the biggest shake up the sport has seen for more than a decade.

Dec 17, 2025 1:00pm UTC

F1 Correspondent & PresenterLawrence Barretto
From 2026, Formula 1 will feature a revised aerodynamic package that will deliver new-look cars alongside an overhauled set of power unit rules that are engaging for existing manufacturers Ferrari and Mercedes, attractive to newcomers Red Bull Powertrains – who have teamed up with Ford – Audi and General Motors (the latter launching their power unit in 2029) and attractive enough to bring Honda back to the party…

Oooooh, interesting. Okay, you've succeeded in getting my attention. How big are we talking?

Well, let's kick off with the aero rules. For starters, the cars will be shorter, narrower, lighter and nimbler. The wheelbase has been shortened, which in theory means they should be more responsive through corners. The cars will still be fitted with 18-inch Pirelli tyres – but they are narrower. That'll cut drag – and trim off yet more weight. Oh – and those little arches above the front tyres are no more, which keeps the cars cleaner and yes, you guessed it, lighter again.

I'm liking the sound of this. Keep the good stuff coming…

No problem! There was a lot of focus on the intricate floors underneath the car in the last era of regulations that lasted from 2022-2025. But from 2026, it's all change. It's goodbye long ground-effect tunnels and hello flatter floors with extended diffusers with bigger openings. This will mean less downforce and a higher ride height requirement that should lead to a greater variety of set-ups that suit a greater range of driving styles – thus levelling the playing field.

Good, good. And what's this I hear about different wings?

It seems you're well informed! From next year, both front and rear wings will be simpler, which roughly translates to fewer elements. The rear beam wings will be no more, while at the other end of the car, the front wing will feature narrower elements. The outer sections of the front wing will also offer potential new areas of development and this will certainly be a critical battleground for teams, as the front wing heavily influences the car's overall aero performance.

Lovely stuff. But…

Hang on, I haven't finished with the wing chat yet! The most significant change is the introduction of Active Aero. The cars can adjust the angle of both their front and rear wing elements depending on where they are on track.

In the corners, the flaps stay shut in their default position to maintain grip. On designated straights, drivers can activate low-drag mode, which opens the flaps and flattens the wings, reducing drag and boosting top speed. This is available to every driver, on every lap.

Everything you need to know about the 2026 regulations

So does that mean it's the end for DRS?

Yes, Active Aero means essentially saying goodbye to DRS in its current form, as the rear wing flaps can be opened on every designated straight without the need to be within one second of the car in front. However, being inside one second of your rival does still bring benefits in the form of Overtake Mode.

Oooooh. Overtake Mode. That sounds fun!

Yep, especially because a driver can strategically use it. The mode is just for attack and is triggered when they're within one second of the car in front. That gives them access to extra electrical energy which they can use to deploy to overtake or pressure the driver ahead at a single detection point.

Nice. Do they still have a button to activate maximum power from the engine and battery?

They sure do. This has been renamed the Boost button and drivers can continue to use it in defence as well as overtaking at any point around the lap, providing they have enough charge in their battery, of course. Drivers may use it all at once or spread across the lap, depending on when they have the best chance to attack or where they are most vulnerable.

The drivers will have new tools at their disposal

Great – so that's two helpful tools at the drivers' disposal. Anything else?

Yes, there is. Drivers will oversee their battery recharge. Working with their race engineer, the duo can select from a range of different modes to recharge their battery, from braking and engine energy. That means they have three tools they can use tactically when in the heat of battle.

Let's get stuck into the power unit changes, then. What's happening under the hood?

A lot. While the core is still a 1.6-litre V6 turbo hybrid, the power balance has shifted significantly. From 2026, the internal combustion engine output has been cut while the electric motor has tripled, meaning we've got roughly a 50-50 power split between petrol and electric. This makes the power units more road relevant – and thus more attractive to existing manufacturers Ferrari and Mercedes, newcomers Red Bull Power Trains in partnership with Ford, General Motors (from 2029) and Audi, plus returning suppliers Honda.

That sounds like a fun challenge for the power unit engineers…

You're not wrong – but this is the kind of challenge they love. To power the new hybrid unit, the car's Energy Recovery System (ERS) can now recharge the battery with twice as much energy per lap, through things like recovery under braking or lifting off the throttle at the end of straights. The revamp of the power unit does mean it's the end of the expensive and complex MGU-H (a heat recovery system), which lacked road relevance and added weight.

The cars will run on Advanced Sustainable Fuels in 2026

And what fuel will these new power units be running on?

I'm glad you asked. For the first time ever, Formula 1 power units will be running Advanced Sustainable Fuels, which have been trialled in F2 and F3 in 2025. The fuel is made from cutting-edge sources like carbon capture, municipal waste and non-food biomass – and it is independently certified to meet strict sustainability standards.

That's pretty cool. There's a lot there already – but anything else?

No rule change would be complete without further enhancements to safety – and 2026 is no different. The drivers' survival cell will be subjected to more rigorous tests, while the roll hoop will be strengthened to take 23% more load – that's roughly the weight of nine family cars. The front impact structure design has been tweaked so it now separates in two stages to give drivers greater protection in big crashes where there are secondary impacts after the initial contact.

Out of interest, who shaped these rules?

That would be governing body the FIA in close collaboration with the teams and commercial rights holder Formula 1. The result was a ruleset that has already attracted four manufacturers – which means more competition and more innovation.

The 2026 machines will feature Active Aero

Superb. Do you mind summing it all up for me in a nutshell, please?

Don't mind if I do. Formula 1 cars will continue to be fast, to be cool, to be awe-inspiring to watch. But from 2026, they will be more challenging for the teams and the drivers. They'll have to deal with new tech and tighter rules – while managing a greater set of tools to attack or defend that can make or break their ultimate performance and finishing position.

With less downforce and tighter control over turbulent air, following a car through a corner should be easier – while getting the best out of the car should be a greater challenger for the drivers. This is the future of Formula 1, powered by advanced sustainable fuel and smarter energy use. Bring. It. On.

EXPLAINED: The key terms for Formula 1's new-for-2026 rules

New rules – complete with new terminology – are coming for 2026. But don't fret – we've got you covered for all the jargon you're going to be hearing throughout the season.

Dec 17, 2025 1:00pm UTCF1 Correspondent & PresenterLawrence Barretto

With Formula 1 introducing new aerodynamic and power unit rules for 2026, here's a rundown of the key new terminology you'll be hearing when the new season gets underway…

Boost Button

As has been the case for several years now, drivers can press a button at any point over the course of a lap to activate energy deployment. From 2026, this will be known as the Boost Button.

When engaged, it will trigger a change in power unit power settings, either returning to maximum power or a profile configured by the team as per their personal choice. The boost could, in theory, either help them defend from a car behind or attack a car ahead providing they have saved enough charge. It can be used all at once or spread across the lap, depending on when they have the best chance to attack or where they are most vulnerable.

Overtake Mode

This is new-for-2026. It’s a performance aid designed to deliver closer racing and boost overtaking opportunities. It can only be activated when a driver is within a second of the car in front at the detection point, which is nominally the final corner.

It allows a driver to recharge an extra +0.5MJ (megajoules) and generate an additional electrical power profile to allow them to sustain a higher speed for a longer period. It can only be used on the following lap. Its use will be most effective on longer straights as the delta speed it will deliver should have a greater effect on closing the distance between the two competing cars.

Everything you need to know about the 2026 regulations

Recharge

Cars will harvest energy to charge the battery when braking, on part throttle, when lifting off (when a driver lifts off the throttle early – often referred to as lift and coast) or when ‘super clipping’ (when some harvesting happens at the end of the straight when a car is still at full throttle – although this depends on the particular engine map being used, the circuit profile, and the overall Recharge energy allowance for that circuit, which varies as per the regulations).

Most of the time, the Recharge will be automated by use of selectable Recharge maps and targets so braking, part throttle and super clip will all be automated functions controlled by the ECU. The only Recharge mode the driver will have direct control of will be lift-off regen, whereby if the driver lifts off, then they can Recharge. However, doing this will disable the Active devices as well. In contrast, super clipping is still at full throttle and therefore the Active Aero will still be ‘open’.

Drivers will have new tools to play within 2026

Active Aero

For the first time, F1 cars will dynamically adjust the angle of both their front and rear wings depending on where they are on the circuit (though it is true to say F1 did have adjustable front wings in 2009 and obviously the DRS rear wing from 2011).

In the corners, the flaps will be in their default 'closed' position to maintain downforce. They will move to their 'open' position to engage a low-drag mode, flattening the wings to reduce drag and increase top speed.

The rear wings can open on defined straights as with DRS now, though there will be more of them per circuit – and you don't need to be inside one second of the car in front to open them.

The 2026 cars will feature Active Aero

The drivers will manually ‘open’ the wings for each zone. Similar to DRS now, drivers can manually 'close' it or the wings will automatically close if they brake or lift off the throttle. There is an additional auto turn-off feature for 2026, whereby for certain zones there is also a lap distance turn-off to ensure that drivers can’t try to take certain corners with the wings ‘open’.

That will prevent incidents like Jack Doohan's off at T1 in Suzuka in 2025.

GALLERY: New look, new tech, new rules – F1 reveals renders of the innovative 2026 car

Take a detailed look at the next generation of sleek, nimble F1 cars that will debut in 2026.

Dec 17, 2025 3:25pm UTC8

An overhaul of Formula 1's technical regulations is coming for the 2026 season, heralding a new era for the sport that will showcase both racing and real-world innovation like never before.

Teams have spent recent years looking ahead to how they can capitalise on the upcoming rule changes, in which existing manufacturers like Ferrari and Mercedes will face new challengers in Red Bull Powertrains (working with Ford), Audi and Honda.

With sweeping adjustments being made to power units, aerodynamic design and safety features, it's harder than ever to predict the competitive order before teams make their 2026 debut at pre-season testing.

The cars will be narrower and lighter, and the sport bids farewell to long ground-effect tunnels, as well as DRS, or the Drag Reduction System. Unlocking more set-up options and making it easier to follow rival cars, the new regulations give teams and drivers alike plenty to sink their teeth into over the winter break and beyond.

2026 REGULATIONS EXPLAINED: All you need to know about F1's new power units

Formula 1 has all-new power units for 2026. A lot of numbers and stats are being thrown around – but what do they really mean?

Jan 14, 2026 3:33pm UTCSpecial ContributorMatt Youson

Twelve years is a good run for a set of engine regulations. Having made their debut in 2014, F1's first genuine generation of hybrid power units took the sport to new levels. They were the most reliable and most economical F1 engines of all time – but also powered the fastest F1 cars of all time – and they were arguably the most closely-matched F1 engines of all time, with each manufacturer scoring wins in double-figures.

But all things come to pass. A new generation of power units takes over for 2026...

Changes have been made to ensure we see more wheel-to-wheel racing… while at the same time, these new engines become simpler, more road relevant and thus more attractive to new entrants. The hybrid element of the power unit becomes more significant, while the sport also switches over to sustainable fuel blends. There's a lot of new terms to assimilate but, just like in 2014, in a few months it'll all seem quite straightforward.

Until then, however, here's our Q&A on the runners and riders and rules and regs for 2026 and beyond.

Not just the cars, but also the power units within them, will be changing in 2026

So, exactly how many power unit manufacturers does F1 have?

The first goal for these regs was to attract more engine builders, and that's been a resounding success. F1 in 2026 has five power unit manufacturers, of which one is new and two are new-ish. Mercedes and Ferrari are the continuity names, and will be powering the bulk of the field. Mercedes have their works team, Williams, a new customer in Alpine and World Champions McLaren.

Ferrari will power the works Ferrari, plus Haas and new F1 entry Cadillac. On top of that, we have a new collaboration between Red Bull and Ford that will power the two Red Bull teams; Honda return as a fully-fledged manufacturer with Aston Martin and, having taken over Sauber, the new Audi team of course have a new Audi power unit.

Remind me, why are we calling it a power unit rather than an engine?

Feel free to call it an engine! The term power unit has been used for the last decade-and-a-bit to help differentiate between the Internal Combustion Engine (ICE), which sits at the heart of the unit, and the wider system that also includes the Turbocharger (TC), battery (ES – for Energy Store), Control Electronics (CE) and the Motor-Generator Unit (MGU-K) that harvests energy but also feeds electrical power into the drivetrain.

Everything you need to know about the 2026 regulations

Is the whole hybrid system new, or are some parts of the original hybrid being retained?

A little bit from column A, a little from column B. The philosophy of F1 operating with a cutting-edge hybrid power unit, some distance ahead of the technology on the road remains, but the architecture of that power unit is changing to make it more road-relevant, while also pushing the limits of what the hybrid part of the hybrid power unit is capable of delivering.

The new unit retains the 1.6-litre turbocharged V6 engine (with a few tweaks) but deletes the MGU-H. The outgoing power unit featured two energy recovery systems: the MGU-K harvested energy under braking (the 'K' stands for Kinetic) while an MGU-H ('H' for Heat) harnessed the flow of exhaust gases.

The latter was deemed surplus to requirements. While very effective, it was a complicated system. That wouldn't necessarily be a deterrent for F1, but in the 18 years since the original F1 hybrids were sketched out, the technology hasn't been adopted by road car manufacturers. Deleting the MGU-H and concentrating on the MGU-K – similar technology to which is present in road cars – makes the new unit less complicated and thus more attractive to manufacturers.

As well as the aim of promoting closer racing, F1's new power units are simpler and more road relevant

Does deleting the MGU-H mean F1 cars will be less reliant on electrical energy?

No. The opposite. The old units had around 20 per cent of their power supplied by the electrical system. The goal for 2026 is to increase this to around 50 per cent. The new MGU-K is nearly three times as powerful as its predecessor, delivering 350kW to the rear wheels, up from 120kW. Likewise, it will recover energy at a greatly increased rate, when the car is braking, coasting and also on throttle.

I hear talk of sustainable fuel. What does that entail?

This is part of F1's commitment to reaching net zero by 2030. All teams will use Advanced Sustainable Fuels this year, and these were successfully trialled by F2 and F3 in 2025.

The fuel is made from cutting-edge sources like carbon capture, municipal waste and non-food biomass – and it is independently certified to meet strict sustainability standards.

Several new tools will be available to the drivers to help with overtaking and defending

Words like 'boost' and 'recharge' are flying around. What do they mean?

These F1 cars will harvest energy to charge the battery when braking, on part throttle, when lifting off (when a driver lifts off the throttle early – often referred to as lift and coast) or when 'super clipping' (when some harvesting happens at the end of the straight when a car is still at full throttle). For the most part, this will be automated, handled by engine maps prepared in advance – but the drivers have the ability to override the automatics.

The only Recharge mode the driver will have direct control of will be lift-off regen, whereby if the driver lifts off, then they can Recharge. However, doing this will disable the Active Aero devices on the car as well. In contrast, super clipping is still at full throttle and therefore the Active Aero will still be 'open'.

Regarding 'boost', for several years drivers have been able to press a button at any point over the course of a lap to activate energy deployment. From 2026, this will now be known as the Boost Button.

When engaged, it will trigger a change in power unit power settings, either returning to maximum power or a profile configured by the team as per their personal choice. The boost could, in theory, either help them defend from a car behind or attack a car ahead providing they have saved enough charge. It can be used all at once or spread across the lap, depending on when they have the best chance to attack or where they are most vulnerable.

There is also an 'overtake' mode. What does this do?

This mode is new, and replaces the Drag Reduction System (DRS) used in F1 since 2011. DRS existed to compensate on the straight for the downforce lost by a following car through the corners.

Overtake mode will do the same. It can only be activated when a driver is within a second of the car in front at the detection point, which is nominally the final corner.

It allows a driver to recharge an extra +0.5MJ (megajoules) and generate an additional electrical power profile to allow them to sustain a higher speed for a longer period. It can only be used on the following lap. Its use will be most effective on longer straights as the delta speed it will deliver should have a greater effect on closing the distance between the two competing cars.

2026 REGULATIONS EXPLAINED: All you need to know about F1's new aerodynamics

F1 goes into the 2026 season with an entirely new chassis philosophy. The cars will be lighter, more nimble, and better able to overtake… but what does it all really mean for racing?

Jan 27, 2026 2:33pm UTCSpecial ContributorMatt Youson

Every few seasons, Formula 1 hits the reset button. While minor tweaks to the technical regulations are common between the seasons, sometimes a bigger change is required, and the rules of F1 are rewritten in a more substantial way. When this happens, we'll get a new engine formula, or perhaps a new aerodynamic concept. 2026 is unusual in that we're having both at the same time.

'Unusual' really doesn't do it justice. It's unprecedented – but it's being done for a very good reason. F1's power units are having their first major refresh for a dozen years, and there will be a new aerodynamic concept to complement them, which leads to a shakeup of the pecking order with everyone starting from scratch.

It's also a car designed to be safer – because F1 never misses an opportunity to use the experience gained over the last few years to build something stronger and better able to deal with whatever situations it encounters.

It is a *lot* of change, and the new and the novel tend to swamp everyone in jargon, but Formula 1 will still be delivering heart-in-the-mouth action… just maybe a bit more of it.

The front wing has been simplified with fewer elements

Will the cars look different?

They will look different – but the sort of different that tends to get forgotten after a few months of the new season.

The main points to notice are that the front wing is simplified with fewer elements (the razor-like blade cascade) but more detail around the endplates. The 'eyebrow' winglets above the front wheels are gone, and there are bargeboards behind the front wheels.

At the rear of the car, the beam wings (that sit below the exhaust) are gone. Less noticeable – but perhaps of greater importance – the underside of the car is very, very different. The ground-effect generating Venturi tunnels carved through the floor are gone, replaced with a flat(ter) design, and a much larger diffuser at the rear of the car.

Does removing the tunnels mean the ground-effect era is over?

Yes. The tunnels under the car encouraged high-speed airflow, which in turn generates a huge amount of grip by sucking the car down to the surface. These forces don't disappear with a flat floor but they are greatly diminished, and these new cars will be finding downforce in other areas.

The ground-effect era succeeded in many of its aims but as the teams developed their cars and recaptured more of their out-washing characteristics (more on that below), it got harder and harder to get close to the car in front. The new regs should promote closer racing.

Everything you need to know about the 2026 regulations

How does the shape of the car promote closer racing?

Glad you asked! The front wheels of an F1 car punch a big hole in the air and create a lot of turbulence. Turbulence is the enemy of good aerodynamic performance. Generations of F1 designers have sought to push that turbulent air out to the sides of the car – outwash – which allows smoother air to infill and make the aerodynamic surfaces wrapping the rest of the car work much more effectively.

The problem with out-washing is that it creates a mass of roiling turbulent air behind the car, making it very difficult for a following car to stay close. These regs, mandating simpler front wings and in-washing bargeboards, are a robust solution designed to prevent out-washing.

We hear the car is to be lighter and more nimble. How? And why?

The minimum weight of an F1 car has been steadily creeping up, adding around 200kg over the last 20 years. More safety features and ferocious crash tests, combined with the extra weight of hybrid systems, have pushed it ever higher. It's created a situation where the cars were very fast, very safe… but not quite as nimble as their predecessors.

So, for this year, the size of the cars has been reduced, with a wheelbase (the distance between the two axles) that is 200mm shorter, and a floor that is 100mm narrower. The tyres are also slimmer, with the tread width reduced by 25mm at the front and 30mm at the rear. Their total diameter also shrinks, by 15mm at the front and 30mm at the rear.

The regs reflect this, with the minimum weight reduced from 800kg to 768kg. There's no guarantee the teams will be able to reach that minimum – indeed, reducing weight is expected to be one of the key engineering challenges.

Bargeboards behind the front wheels return to help prevent out-washing

Does this mean the car is less safe?

No. None of the crash tests have been diluted. The cars will have to pass homologation tests at least as stringent as those faced last year. In several instances, however, the tests will be tougher.

There's a new crash test for the roll hoop (the structure on top of the car), the load for which rises from 16g to 20g, while the nose cone now features a two-stage impact structure. This is to deal with the phenomenon of secondary impacts: it's not uncommon to see the nose cone crumple as designed to absorb a heavy impact, but then continue to spin and have a secondary impact, with the crash structure already detached. The new version provides greater protection.

On the subject of the nose, the front wing now has an 'active' component. What's active about it?

Both the front and rear wings have active – i.e. moveable – elements. These operate like a Venetian blind. When the blades are 'closed' they are providing downforce, but also creating a lot of drag. When they 'open', both drag and downforce are reduced.

It's a best-of-both words situation: downforce for higher speeds through the corners (Corner Mode); reduced drag for higher speeds on the straights (Straight Mode).

The underside of the car is very different, with a flatter floor and a much larger rear diffuser

Since 2011, the rear wing has had an active element within the Drag Reduction System (DRS). It's use during a Grand Prix was restricted, only chasing cars within a second of the car in front were allowed to use it, and then only on specified straights. The new system is useable by all of the cars, all of the time.

The reason the front wing will now also realign, is to better trim the car and provide more stability, in harmony with the movement of the rear wing. It’s common to adjust the front wing manually, in the garage and during pit-stops, to better balance the car, and F1 tried a driver-adjustable front wing in 2009, without much success.

2026 REGULATIONS EXPLAINED: All you need to know about F1's Advanced Sustainable Fuels

Formula 1 is introducing Advanced Sustainable Fuels in 2026 as part of the biggest shake up the sport has seen for more than a decade. It’s a huge technical change… but one where success will be measured by nobody noticing.

Jan 30, 2026 4:22pm UTC Special Contributor Matt Youson

A great deal in Formula 1 is changing for 2026, with new power units, a new aerodynamic philosophy and new sporting regulations. The switch to Advanced Sustainable Fuels isn’t perhaps garnering the same volume of headlines, but it has the potential to be just as ground-breaking.

With redesigned power units scrapping for every last ounce of energy, F1’s fuel suppliers are going to be waging their own private battle over the next few years, competing to see who can push this new technology the furthest.

Why is F1 doing this now?

F1 has a commitment to becoming Net Zero by 2030. It’s a hugely ambitious goal, and takes in many measures, large and small, from powering the garages, paddock and broadcast centre with solar and used vegetable oil, to banning the use of single-use plastics.

Operating the cars with Advanced Sustainable Fuel is a small part of the whole – but one which has a tremendous resonance. Formula 1 sets the tone: it inspires and leads. Switching over to use sustainable fuels in F1 power units is a potent statement.

Formula 1 cars will run on cutting-edge Advanced Sustainable Fuels in 2026

What is an Advanced Sustainable Fuel?

For the purposes of F1, it’s an e-fuel made from cutting-edge sources like carbon capture (taking CO2 directly from the air or industrial emissions), municipal waste and non-food biomass, or a mix of the above.

Chemists can cherry-pick the molecules they want for their blend – and it's independently certified to meet strict sustainability standards. Nothing in the fuel comes from crude oil.

Are these genuinely good for the planet?

Producing the fuels can be quite energy-intensive, though the more the processes can be scaled-up, the better the picture looks. The spec for F1 is that the process must be powered by renewables,

rather than anything generated using fossil-fuelled electricity, and it is independently certified to meet strict sustainability standards.

For a fuel created from biomass, for example, it has to be produced using waste biomass rather than crops grown on arable land that would otherwise be used to grow food.

Does a sustainable fuel require a different sort of engine?

No. These are 'drop-in' fuels, meaning they're designed to replace fossil fuel equivalents without requiring any adjustment to the engine. They would work in road cars, and we've also seen plenty of demonstrations on track. Sebastian Vettel and his 'Race without Trace' project demonstrated sustainable fuels by powering a Nigel Mansell Williams FW14B, an Ayrton Senna McLaren MP4/8, and his own Red Bull RB9.

Everything you need to know about the 2026 regulations

Of course, every fuel raced in F1 is tailored for the specific era in which it is used, and designed for the engine in which it will race. With the previous generation of engines having been used for 12 years, fuel technology, while competitive, had matured to the point where it hasn't been a huge differentiator.

With a new generation of power units being introduced in 2026, designing fuel for it would have been a vital part of the technical race, even without the switch away from fossil sources. Going down the sustainable route makes this an incredible challenge for the fuel manufacturers, and will absolutely have an impact on the competitive position.

Do sustainable fuels produce as much power as their fossil alternatives?

We really don't know! In terms of bang-for-buck, fossil fuels are exceptionally calorific, and it would be a big ask for the sustainable alternatives to have that level of energy density from day one.

What we are expecting is something very close to the fossil fuel level of performance, that develops rapidly over the next few years. It's going to be impossible to really know, given the blends will be different this year (F1 teams have their own fuel suppliers). Plus, there's the uprated hybrid system providing almost 50 per cent of the power from recovered electrical energy.

Of course, F1 isn't going into this entirely blind. Aramco produced the sustainable fuel used in Formula 2 and Formula 3 last year, and that had no adverse impact on performance.

Aramco produced the Advanced Sustainable Fuel used in both Formula 2 and Formula 3 last year

Hasn't F1 been using a sustainable fuel already too?

Yes – but not to this extent. In the last few years, F1 has been using E10 blends – i.e. fuels mixed with a 10 per cent Ethanol component. That's provided some useful learning, but this is an altogether bigger step.

Beyond the track, however, yes, lots of the background infrastructure in F1 already uses a sustainable fuel source. Some teams use e-fuels and biofuels for their generators and forklifts, and logistics partner DHL use biofuel in their trucks.

At the Austrian Grand Prix we've even seen the whole paddock, pit-building and Broadcast Centre powered by a combination of solar panels and hydrotreated vegetable oil (HVO), delivering an estimated 2.5MWh, in a very successful trial.

There's been some talk of fuel use being measured differently this year…

It is. Since 2014, F1 cars have been limited in the amount of fuel they can use, both in the fuel flow at any given moment in the race but also across the race as a whole.

That still exists, but for this year, the emphasis moves off the quantity of fuel used (measured in kilograms per hour) and onto the energy density of that fuel (measured in megajoules per hour), with a hard limit of 3000MJ/h.

The maths behind this is complex, but the car's electronic brain will do the necessary calculations, based on fuel samples examined and certified in advance by an independent body. As has been the case for a while, the random fuel samples taken at the track by the scrutineers will be examined to ensure the 'fingerprint' of live samples matches the reference on file.

F1 has a commitment to becoming Net Zero by 2030

Is the sustainable fuel technology something we're going to see transferred to the road?

This is certainly the intention. For over 30 years, F1 had insisted race cars run on something suitable for the road.

It may be bespoke, and tailored for the demands of a race car, but it contains the same ingredients and has a similar octane rating to what you would find on the garage forecourt (i.e. no exotic 'rocket fuels').

The new fuels are the same: the chemistry done in the lab for the F1 projects, whether that's distillation profiles, detergents or octane numbers, will have direct relevance to the road cars and freight fleets of the future.